



SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2024 – November 2024		
Maximum Marks: 30	Examination: In-Semester Examination	Duration : 1.15 hrs
Programme code: 54 Programme: B.Tech Computer Engineering (Honours in Data Science & Analytics)	Class: LY	Semester: VII (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: COMP - Honours (DSA)	
Course Code: 116h54C701	Name of the Course: Advanced Machine Learning	

Question No.		Max. Marks	CO Mapped	BT Level
Q1	Consider the following linear model where the loss function is defined as: $L(w) = \frac{1}{2}(y - (w_1x_1 + w_2x_2))^2$ Given: • Target value: $y = 10$ • Features: $x_1 = 3, x_2 = 4$ • Initial weights: $w_1 = 0.5, w_2 = 0.2$ • Learning rate: $\alpha = 0.01$ Answer following: i) Compute the gradients $\frac{\partial L}{\partial w_1}$ and $\frac{\partial L}{\partial w_2}$. ii) Using the computed gradients, update the weights w_1 and w_2 for the first iteration. iii) Repeat the gradient calculations and weight updates using the updated weights from the first iteration. Show all calculations for the second iteration. OR Consider a Multi-Layer Perceptron (MLP) to perform image classification on a dataset of 28×28 grayscale images (flattened into 784-dimensional vectors), into one of 10 possible classes (digits 0-9) for a following network architecture: • Input Layer: 784 neurons • Hidden Layer 1: 128 neurons • Hidden Layer 2: 64 neurons • Hidden Layer 3: 32 neurons • Output Layer: 10 neurons with Softmax activation Answer following: a) Draw the architecture of the network and Compute the total	10	CO1 & CO2	AP & AN

	number of parameters in the network. b) Explain the role and choice of activation functions (ReLU in hidden layers and Softmax in the output layer). c) Outline the process of computing the network's output for an input vector and provide a brief overview of the back-propagation process for this network. Describe how gradients of the loss function are computed and how they are used to update weights and biases. Note: For answer b and c, explain with proper mathematical equations.																							
Q2	Draw and describe the architecture of CNN - LeNet-5. Include details about the number of layers, types of layers (convolutional, pooling, fully connected), and activation functions used. Calculate the output size after each layer in the LeNet-5 architecture, including convolutional and pooling layers.	10	CO3	RE, UN & AP																				
Q3	Answer Any Two: a) Explain Forward Propagation in RNN. b) Consider a machine learning application for predicting stock prices. Identify the type of learning and algorithm you would use and justify your choice. c) Consider you are building multiple models to solve a binary classification problem. For each model, you compute the Precision and Recall on both the training set and the validation set. The results are shown below: <table border="1"> <thead> <tr> <th>Model</th><th>Precision (Training)</th><th>Recall (Training)</th><th>Precision (Validation)</th><th>Recall (Validation)</th></tr> </thead> <tbody> <tr> <td>P</td><td>95%</td><td>92%</td><td>70%</td><td>65%</td></tr> <tr> <td>Q</td><td>80%</td><td>75%</td><td>78%</td><td>74%</td></tr> <tr> <td>R</td><td>85%</td><td>88%</td><td>84%</td><td>85%</td></tr> </tbody> </table> Answer the following questions: i. Which model is likely overfitting the data? ii. Which model is likely underfitting the data? iii. If you had to select one of these models to deploy, which one would you choose and why?	Model	Precision (Training)	Recall (Training)	Precision (Validation)	Recall (Validation)	P	95%	92%	70%	65%	Q	80%	75%	78%	74%	R	85%	88%	84%	85%	10 (05 + 05)	CO1, CO2 & CO3	UN, AP & AN
Model	Precision (Training)	Recall (Training)	Precision (Validation)	Recall (Validation)																				
P	95%	92%	70%	65%																				
Q	80%	75%	78%	74%																				
R	85%	88%	84%	85%																				